Marketing Campaign Performance & Sales Analysis Report

1. Introduction

This project focuses on analyzing marketing campaign data to understand how different advertising methods influence product sales. The primary objective is to transform raw marketing data into meaningful insights using Microsoft Excel and Power BI.

The project emphasizes not only data visualization, but also the complete data preparation process, including cleaning, structuring, and transforming data into an analysis-ready format before performing analysis.

2. Project Objectives

The objectives of this project are:

- To convert raw marketing data into analysis-ready format
- To study the relationship between marketing spend and sales
- To identify high-performing advertising channels
- To evaluate cost efficiency
- To design a professional dashboard for business users
- To present insights in a clear and understandable way

3. Dataset Overview

The dataset contains information related to:

- Product Sold
- Traditional, Digital, Partnership, and Trust Spend
- Performance Channel
- Support Channel
- Cost Per Sale

- Total Spend

Initially, the dataset was not structured for direct business analysis. Therefore, several transformations were required.

4. Tools Used

i. Microsoft Excel – Initial cleaning and understanding
ii. Power BI – Data modeling, DAX calculations, and dashboard design

5. Data Transformation & Preparation Process

Before analysis, the raw dataset was converted into analytical format. This was the most important stage of the project.

5.1 Data Type Correction

First, all columns were checked for correct data types.

- Product_Sold → Converted to Whole Number
- Spend Columns → Converted to Decimal Number
- Category Columns → Converted to Text

This step was required to avoid calculation and DAX errors.

5.2 Grouping Spend Methods into Categories

The original dataset contained multiple spend methods. For better analysis, they were grouped into four main categories:

Category	Methods Included
Traditional	TV, Billboard
Digital	Google Ads, Social Media
Partnership	Affiliate marketing
Trust	Influencer marketing

Why This Was Done:

- Raw methods were too detailed for strategic analysis
- Businesses usually evaluate at category level
- Helps compare major marketing strategies

This grouping simplified the dataset and improved decision-making relevance.

5.3 Grouping Categories into Channels

Next, categories were grouped into two channels:

Channel Type	Categories
Performance Channel	Traditional, Digital
Support Channel	Partnership, Trust

Why This Was Done:

- Performance channels directly drive sales
- Support channels assist indirectly
- Helps compare core vs supporting marketing efforts

This made it easier to analyze campaign structure.

5.4 Creating Product Sold Ranges (Buckets)

To understand sales patterns, Product_Sold was converted into ranges:

Range Name	Values
0–5000	Low Sales
5000–7000	Medium-Low
7000–9000	Medium-High
9000+	High Sales

Why Buckets Were Created:

- Raw numbers are difficult to compare visually
- Buckets show growth stages
- Helps analyze performance at different business levels

Why These Ranges Were Selected:

- Based on data distribution
- Balanced number of records
- Avoided extreme imbalance
- Covered business-relevant sales levels

5.5 Creating Bucket Order Column

Since ranges are text, Power BI sorts them alphabetically.

So a Bucket Order column was created:

Range	Order
0–5000	1
5000–7000	2
7000–9000	3
9000+	4

This was used to sort visuals correctly.

5.6 Calculating Total Spend

Total Spend was calculated as:

- Total Spend = Traditional + Digital + Partnership + Trust

Why:

- To know overall marketing investment
- Used for KPI and ROI analysis
- Required for management reporting

5.7 Calculating Cost Per Sale

Cost Per Sale was calculated as:

- Cost Per Sale = Total Spend / Product Sold

Why This Was Important:

- Shows efficiency of marketing
- Measures how much is spent to sell one unit
- Helps compare campaigns fairly
- Used to evaluate ROI indirectly

Without this metric, efficiency cannot be measured.

5.8 Why Averages Were Used Instead of Totals

In many visuals, Average Spend was used instead of Total Spend.

Reason:

The dataset has uneven bucket sizes.

Example:

- 0–5000 → ~35 records
- 5000–7000 → ~109 records
- 7000–9000 → ~125 records
- 9000+ → ~31 records

If totals were used:

- Buckets with more records would automatically look better.

This creates misleading insights.

Using averages:

- Removes record count bias
- Allows fair comparison
- Shows true performance

Therefore, averages were preferred.

6. Exploratory Data Analysis (EDA)

After transformation, EDA was performed.

Activities:

- Analyzed spend vs sales
- Compared categories across buckets
- Studied cost efficiency
- Checked distribution patterns
- Verified channel performance

EDA helped validate assumptions before visualization.

7. Dashboard Design & Structure

The dashboard was designed on a single page for easy understanding.

7.1 KPI Section

Top-left cards show:

- Total Marketing Spend
- Average Cost Per Sale
- Total Units Sold

Purpose: Quick performance snapshot.

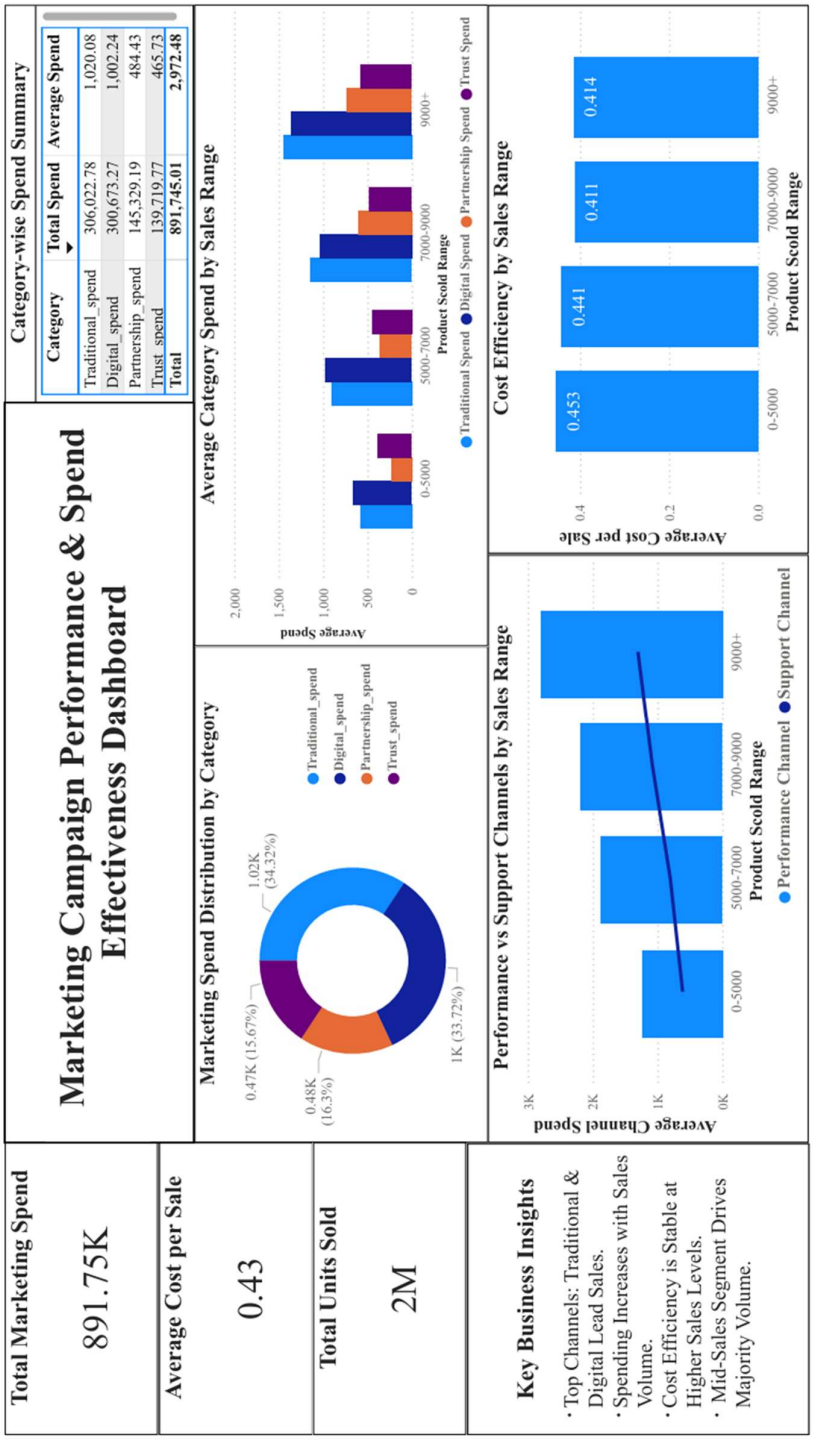
7.2 Main Visual Section

Visual	Purpose
Donut Chart	Category spend share
Bar Chart	Category vs sales range
Combo Chart	Channel comparison
Column Chart	Cost efficiency
Table	Detailed spend summary

Each visual answers a specific business question.

7.3 Dashboard Placement

After analysis, the final dashboard is inserted in the report to visually support findings.



The dashboard was placed after analysis to visually represent the key findings and support business decision-making.

8. Key Business Insights

i. Traditional and Digital channels generate highest sales
ii. Higher sales require higher investment
iii. Cost per sale remains stable at high volumes
iv. Mid-sales segment (5000–9000) drives majority revenue
v. Support channels need optimization

9. Business Recommendations

- Increase investment in Traditional & Digital
- Improve efficiency of Trust & Partnership
- Focus on mid-sales growth segment
- Track cost per sale regularly
- Reallocate underperforming budgets

10. Challenges Faced

- DAX syntax errors
- Circular dependency issues
- Sorting problems
- Data imbalance
- Understanding business logic

These improved my problem-solving skills.

11. Learning Outcomes

Through this project, I learned:

- Data transformation techniques
- Power BI modeling
- DAX calculations
- Business-oriented analysis
- Dashboard storytelling

12. Conclusion

This project demonstrates how raw marketing data can be transformed into meaningful business insights through systematic preprocessing, analysis, and visualization.

By grouping data, creating meaningful metrics, and using appropriate aggregations, a decision-support dashboard was developed. This project has helped me improve both my technical and analytical skills.